

FUNDAMENTALS OF DATA MINING

Lecture

Association Rule – Apriori and FP-Growth

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Apriori Vs FP-Growth

- Apriori: Uses a generate-and-test approach – generate candidate itemsets and test if they are frequent
 - ▣ Generation of candidate itemsets is expensive (both time and space)
 - ▣ Support counting is expensive
- FP-Growth: allows frequent itemset discovery without candidate itemset generation.

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FP-growth Algorithm

- Use a compressed representation of the database using an **FP-tree**
- Once an FP-tree has been constructed, it uses a recursive divide-and-conquer approach to mine the frequent itemsets

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Mining Frequent Patterns using FP-Tree

- Key Steps
 - ▣ Build the FP-Tree
 - ▣ Build the Header table
 - ▣ Create conditional pattern bases
 - ▣ Create conditional FP-Tree
 - ▣ Enumeration of all frequent patterns in each conditional FP-Tree

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FP-Tree Construction

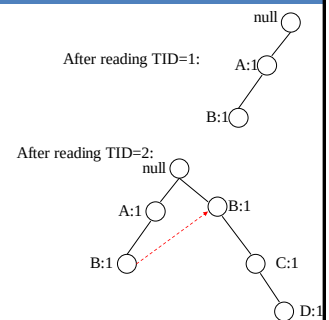
- FP-Tree is constructed using 2 passes over the data-set:
 - ▣ Pass 1:
 - Scan data and find support for each item.
 - Discard infrequent items.
 - Sort frequent items in *decreasing order* based on their support.
 - For our example: A; B; C; D; E
 - Use this order when building the FP-Tree, so common prefixes can be shared.
 - ▣ Pass 2: Construct the FP-Tree

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FP-Tree Construction

- After removing infrequent items (Sup > 1)
- Sort frequent items in *decreasing order* based on their support

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
4	{A,D,E}
5	{A,B,C}
6	{A,B,C,D}
7	{B,C}
8	{A,B,C}
9	{A,B,D}
10	{B,C,E}

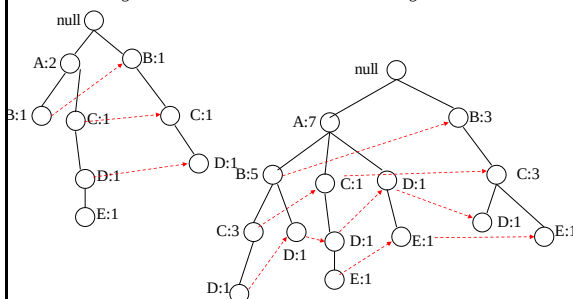


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FP-Tree Construction

After reading TID=3:

After reading TID=10:



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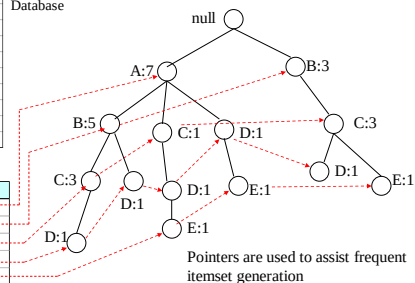
FP-Tree Construction

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
4	{A,D,E}
5	{A,B,C}
6	{A,B,C,D}
7	{B,C}
8	{A,B,C}
9	{A,B,D}
10	{B,C,E}

Transaction Database

Header table

Item	Pointer
A	
B	
C	
D	
E	

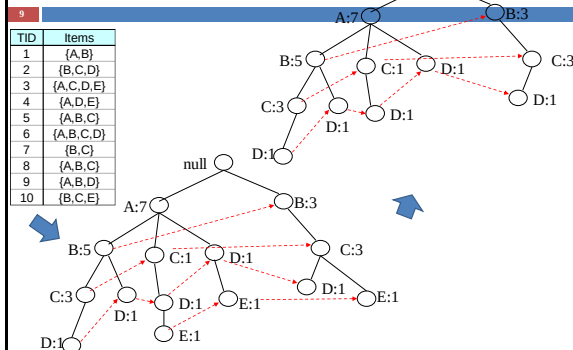


Pointers are used to assist frequent itemset generation

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Prefix path sub-tree for D

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
4	{A,D,E}
5	{A,B,C}
6	{A,B,C,D}
7	{B,C}
8	{A,B,C}
9	{A,B,D}
10	{B,C,E}



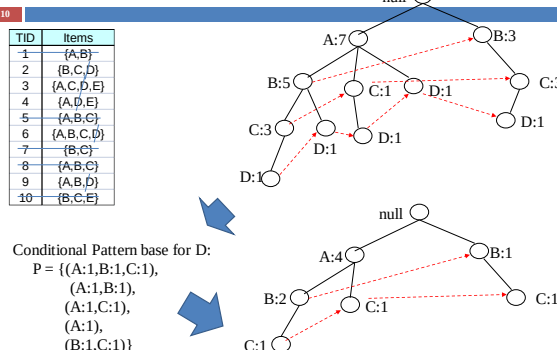
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Conditional FP-Tree for D

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
4	{A,D,E}
5	{A,B,C}
6	{A,B,C,D}
7	{B,C}
8	{A,B,C}
9	{A,B,D}
10	{B,C,E}

Conditional Pattern base for D:

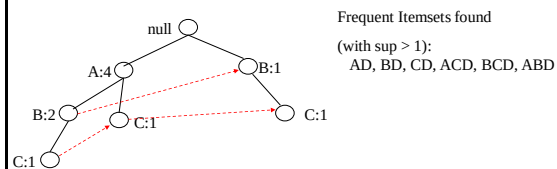
P = {(A:1,B:1,C:1),
(A:1,B:1),
(A:1,C:1),
(A:1),
(B:1,C:1)}



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Frequent Itemsets

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Frequent Itemsets found

(with sup > 1):
AD, BD, CD, ACD, BCD, ABD

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Example 2

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Example of FP-tree

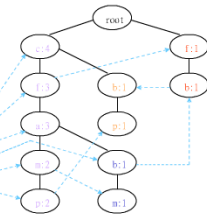
TID	Items Bought	(Ordered) Frequent Items
100	f,a,c,d,g,i,m,p	c,f,a,m,p
200	a,b,c,f,i,m,o	c,f,g,b,m
300	b,f,h,j,o	f,h
400	b,c,k,s,p	c,b,p
500	a,f,c,e,i,p,m,n	c,f,s,m,p

$\xi = 3$

item	head of node-links
c	
f	
a	
b	
m	
p	

Scan of DB derives frequent items:
 $\langle c:4 \rangle, \langle f:4 \rangle, \langle a:3 \rangle, \langle b:3 \rangle, \langle m:3 \rangle, \langle p:3 \rangle$

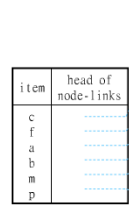
Scan of DB derives FP-tree



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Conditional pattern bases

- Start with the frequent header table.
- Traverse tree by following links from frequent items.
- Accumulate prefix paths to conditional pattern bases.



item	cond. pattern base
c	null
f	c:3
a	cf:3
b	cfa:1, c:l, f:l
m	cfa:2, cfab:l
p	cfam:2, cb:l

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Create conditional FP-Tree

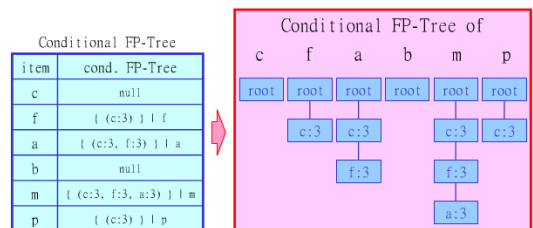
- For every pattern base
 - Accumulate the frequency for each item.
 - Construct as FP-Tree for the frequent items of the pattern base.

item	cond. pattern base
c	null
f	c:3
a	cf:3
b	cfa:1, c:l, f:l
m	cfa:2, cfab:l
p	cfam:2, cb:l

item	cond. FP-Tree
c	null
f	{ (c:3) } f
a	{ (c:3, f:3) } a
b	null
m	{ (c:3, f:3, a:3) } m
p	{ (c:3) } p

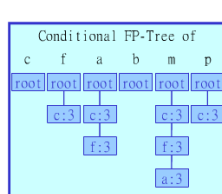
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Conditional FP-Tree



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Enumeration of all frequent patterns



item	Generate all frequent patterns
c	(c:4)
f	(f:4), (cf:3)
a	(a:3), (ca:3), (fa:3), (cfa:3)
b	(b:3)
m	(m:3), (cm:3), (fm:3), (am:3), (cfm:3), (cam:3), (fam:3), (cfam:3)
p	(p:3), (cp:3)

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Example 3

Explain on board

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